

ERIEAU, ON, Canada

WMO: 714650

Lat: 42.257N

Lon: 81.908W

Elev: 584

StdP: 14.39

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.0	8.5	-4.9	4.3	5.0	0.3	5.7	10.1	35.3	21.9	31.7	22.5	10.8	270	0.515

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.9	80.5	73.3	78.8	72.2	77.2	71.1	76.1	78.3	74.6	77.1	73.0	75.8	7.2	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.5	136.3	77.6	73.7	128.2	76.3	72.1	121.2	75.1	40.0	78.7	38.5	77.4	37.1	76.0	81.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
27.8	24.5	21.5	DB	-0.5	85.3	6.5	3.1	-5.2	87.5	-9.0	89.3	-12.6	91.0	-17.3	93.3	(4)
			WB	-1.5	77.1	6.2	2.7	-6.0	79.0	-9.6	80.6	-13.1	82.1	-17.6	84.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.8	25.3	25.9	32.8	43.5	54.3	65.4	70.8	70.8	65.2	54.0	42.9	32.8
	DBStd	17.74	9.18	8.64	7.34	5.84	6.00	5.54	4.41	4.19	6.08	7.06	7.93	8.15
	HDD50	3010	765	676	533	206	24	0	0	0	0	40	233	533
	HDD65	6457	1230	1096	998	646	338	62	6	6	71	344	662	998
	CDD50	2560	0	0	0	9	156	463	644	646	456	165	21	0
	CDD65	533	0	0	0	0	5	75	185	187	76	5	0	0
	CDH74	1414	0	0	0	0	10	123	589	561	131	0	0	0
	CDH80	77	0	0	0	0	1	11	42	21	2	0	0	0
Wind	WSAvg	9.7	12.2	10.8	10.1	10.6	9.0	7.1	6.7	7.0	8.3	10.3	12.2	12.2
Precipitation	PrecAvg	36.1	3.0	2.4	2.4	3.0	3.3	3.1	3.3	2.9	3.2	3.0	3.2	3.0
	PrecMax	47.2	4.6	4.9	4.8	4.9	6.0	6.2	6.5	5.2	8.4	6.1	5.8	5.8
	PrecMin	28.3	1.2	0.7	0.9	1.0	0.9	1.2	1.2	0.5	0.7	0.8	1.2	1.2
	PrecStd	5.6	0.9	1.1	1.1	1.0	1.3	1.3	1.1	1.1	1.8	1.3	1.3	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.8	43.1	49.9	60.2	73.4	81.0	83.8	81.9	80.0	71.0	60.3	52.0
		MCWB	43.3	40.6	44.3	50.5	64.6	70.0	74.8	73.6	73.8	67.7	57.1	50.8
	2%	DB	42.1	40.2	46.2	55.2	68.4	77.1	80.7	80.2	77.3	68.5	58.0	49.1
		MCWB	40.4	37.7	42.6	48.0	61.4	69.8	74.0	73.3	71.8	64.3	54.6	47.5
	5%	DB	39.7	37.9	43.6	52.4	64.7	74.8	79.0	78.5	75.3	66.3	56.0	46.3
		MCWB	37.7	35.8	40.3	46.7	59.2	68.8	72.8	71.8	69.9	62.1	52.4	44.1
	10%	DB	37.0	35.9	41.4	50.3	61.9	72.8	77.2	77.1	73.6	64.1	54.1	43.3
		MCWB	35.2	34.1	38.7	45.4	56.9	67.7	71.3	71.2	68.6	59.6	50.9	40.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.5	41.5	46.4	53.9	66.2	74.8	78.5	78.2	75.5	69.1	58.5	51.0
		MCDB	44.4	42.4	48.0	57.5	70.4	77.1	80.9	79.7	78.0	70.5	59.6	51.7
	2%	WB	40.9	38.2	43.1	50.0	62.8	72.2	76.5	76.1	73.7	64.8	55.7	47.9
		MCDB	41.9	39.6	45.2	52.9	67.1	75.0	78.9	77.9	76.2	67.4	57.1	48.8
	5%	WB	37.9	36.1	40.8	47.9	60.2	70.5	74.7	74.5	71.8	62.9	53.2	44.5
		MCDB	39.2	37.6	42.9	51.0	63.9	73.6	77.3	76.9	74.4	65.7	55.3	46.2
	10%	WB	35.4	34.3	38.9	46.1	58.0	68.7	73.0	72.8	69.5	60.6	51.0	41.2
		MCDB	36.9	35.8	41.2	49.1	61.0	72.0	75.9	75.9	72.8	63.5	53.9	42.9
Mean Daily Temperature Range	5% DB	MDBR	9.6	9.5	9.0	9.6	9.2	8.9	9.2	8.9	9.0	9.3	8.7	8.1
		MCDBR	8.7	9.2	10.9	12.9	12.7	10.8	10.2	9.1	8.5	8.6	9.2	8.2
		MCWBR	8.8	8.5	8.6	9.0	8.2	7.1	7.4	7.4	7.1	7.9	9.1	8.6
	5% WB	MCDBR	8.4	9.0	10.4	11.9	11.2	8.5	8.6	8.0	7.9	8.4	8.9	8.2
MCWBR		8.9	8.7	9.0	9.8	8.4	6.6	7.3	8.0	7.5	8.6	9.3	8.9	
Clear-Sky Solar Irradiance	taub	0.304	0.327	0.347	0.395	0.421	0.440	0.436	0.413	0.380	0.366	0.327	0.305	
	taud	2.392	2.286	2.315	2.267	2.245	2.232	2.270	2.321	2.394	2.403	2.475	2.480	
	Ebn at Noon	266	276	284	277	272	266	266	268	270	259	256	257	
	Edh at Noon	25	33	36	41	43	44	42	38	33	29	23	22	
All-Sky Solar Radiation	RadAvg	459	668	1093	1458	1742	1967	1952	1711	1363	869	534	366	
	RadStd	43	92	111	131	150	128	116	78	117	73	75	31	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (60 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page